Boundedness of Good Minimal Models

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Contents

Classification of curves

One of the central problems in algebraic geometry is to classify algebraic varieties up to birational equivalence.

For smooth projective curves C, the classification is determined by the sign of the canonical divisor K_C:

Type	Condition on K_C	Geometry
Fano	$-K_C$ ample	$\mathbb{P}^1$
Calabi–Yau	$K_C \sim 0$	elliptic curves
General type	K_C ample	genus $g \geq 2$

Surfaces and Kodaira dimension

For a smooth projective surface Y, the **Kodaira dimension** is defined by

$$\kappa(Y) := \dim \mathrm{Im}\big(\phi_{|mK_Y|}\big) \quad (m \gg 0),$$

or equivalently, the growth rate of $h^0(Y, mK_Y)$.

After running the Minimal Model Program $Y \dashrightarrow X$, we obtain either a **minimal model** or a **Mori fiber space**:

$\kappa(Y)$	Output of MMP (on X)
$-\infty$	Mori fiber space: $\mathbb{P}^2$ or $\mathbb{P}^1$-fibration
0	$K_X \sim_{\mathbb{Q}} 0$: K3, Enriques, abelian, bielliptic
1	K_X semiample: elliptic fibration over a curve
2	K_X nef and big: X admits a canonical model Z with K_Z ample

MMP and Abundance

The Minimal Model Program (MMP) and Abundance Conjecture (completely established in dimension ≤ 3) predict that any projective variety Y with mild singularities is birational to either:

- a **Mori fiber space** $X \to Z$, i.e. $\rho(X/Z) = 1$, $-K_X$ is ample over Z (a Fano fibration), and $\dim X > \dim Z$;
- a **good minimal model** X, i.e. K_X is semiample and defines a Calabi–Yau fibration $X \to Z$.

Therefore, **Fano varieties**, **Calabi–Yau varieties**, and **canonically polarized varieties**, together with their iterated **fibrations**, form the basic building blocks in birational geometry.

Moduli spaces: general idea

In algebraic geometry, many classification problems are better understood via **moduli spaces**.

Roughly speaking, a moduli space should parametrize isomorphism classes of varieties with fixed numerical invariants (e.g. dimension, volume).

Expected features:

- points correspond to varieties (up to isomorphism);
- families correspond to morphisms into the moduli space;
- degenerations are included via a suitable compactification.

Moduli of curves

A fundamental guiding example is the moduli space of curves.

Fix $g \geq 2$. The moduli stack $\mathcal{M}_g$ parametrizes smooth projective curves of genus g, and it admits a coarse moduli space M_g.

Key facts:

- M_g is not compact: smooth curves can degenerate.
- The Deligne–Mumford stack $\overline{\mathcal{M}}_g$ provides a natural compactification.
- It admits a coarse moduli space $\overline{M}_g$.
- Boundary points correspond to **stable curves** with nodal singularities.

Why singularities appear

In higher-dimensional birational geometry, singular varieties naturally appear for two reasons:

- **MMP operations:** running MMP typically produce singular varieties.
- **Moduli compactifications:** smooth varieties degenerate to singular ones.

Therefore, one must enlarge the category from smooth varieties to varieties with controlled singularities.

Mild singularities (klt, lc, ϵ-lc)

Let X be a normal variety such that K_X is $\mathbb{Q}$-Cartier.

To measure singularities, we take a log resolution $\pi : X' \to X$ and write

$$K_{X'} = \pi^* K_X + \sum a_i E_i.$$

The numbers a_i measure how singular X is along E_i.

Definitions: X is

- klt (Kawamata log terminal): if all $a_i > -1$
- lc (log canonical): if all $a_i \geq -1$

Remark: If you are not familiar with singularities, you may think of X as smooth.

Boundedness

The notion of **boundedness** naturally arises from the theory of moduli spaces. Roughly speaking, a moduli space requires that all objects in a given class are controlled by a **finite type** parameter space.

Definition (Boundedness)

Let $d \in \mathbb{N}$ and let $\mathcal{P}$ be a set of d-dimensional projective varieties.
We say $\mathcal{P}$ is a ***bounded family*** if there exists $r \in \mathbb{N}$ such that for every $X \in \mathcal{P}$, there exists a **very ample** divisor H on X with $H^d \leq r$.
Equivalently, there exists a projective morphism $\pi : \mathcal{X} \to S$ between **finite type** schemes such that every $X \in \mathcal{P}$ is isomorphic to a fiber $\mathcal{X}_s$ for some $s \in S$.

Example (curves $g \geq 2$):

- K_C is ample,
- $3K_C$ is very ample and embeds $C \hookrightarrow \mathbb{P}^N$.

Moduli of varieties of general type

KSBA: Higher-dimensional analogue of $\overline{\mathcal{M}}_g$

There exists a **projective coarse moduli space** for X such that:

- X is a d-dimensional projective variety,
- X has **slc** (non-normal analogue of lc) singularities,
- K_X is **ample** with fixed volume $(K_X)^d = v$.

This theory has been developed over the past three decades through the contributions of many people.

Key input: boundedness of such varieties.

Boundedness of general type varieties

Theorem (Hacon-McKernan-Xu 2018)

Fix $d \in \mathbb{N}$ and $v \in \mathbb{Q}^{>0}$. Consider the set of varieties X such that

- *X is **slc** of dimension d,*
- *K_X is **ample** with volume $(K_X)^d = v$.*

Then X belongs to a bounded family.

Boundedness of Fano varieties

A celebrated result is Birkar's solution of the **BAB conjecture**.

Theorem (Birkar 2021)

Fix $d \in \mathbb{N}$ and $\epsilon \in \mathbb{Q}^{>0}$. Consider the set of varieties X such that

- *X is ϵ-**lc** of dimension d,*
- *$-K_X$ is **ample**.*

Then X belongs to a bounded family.

Calabi–Yau varieties

For Calabi–Yau varieties, there is in general no natural choice of polarization. As a consequence, they are **not bounded** in the category of algebraic varieties.

For example, projective K3 surfaces and abelian varieties (of any fixed dimension) are not bounded.

It is **widely open** whether strict Calabi–Yau manifolds of dimension $d \geq 3$ are bounded.

Definition (strict Calabi–Yau). A smooth projective variety X is strict Calabi–Yau if

- X is simply connected,
- $K_X \sim 0$,
- $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < \dim X$.

Boundedness of polarized Calabi-Yau varieties

When studying moduli of smooth Calabi–Yau varieties, one may choose a **line bundle** as the polarization.

Since such moduli spaces are **not proper**, one needs to allow singular degenerations. In this setting, the limiting polarization is **no longer Cartier**, but only a $\mathbb{Q}$-Cartier **Weil divisor**.

Theorem (Birkar 2023)

*Let $d \in \mathbb{N}$ and $u \in \mathbb{Q}^{>0}$. Consider Calabi–Yau varieties X and $\mathbb{Q}$-Cartier **Weil** divisors A on X such that*

- *X is **klt** of dimension d,*
- *A is an **ample** divisor on X such that $\mathrm{vol}(A) \leq u$,*

then X belongs to a bounded family.
In particular, the Cartier index of A is bounded.

1 Background

2 Main results
- Faiture of boundedness for polarized lc good minimal models
- Boundedness of polarized klt good minimal models
- Moduli of polarized klt good minimal models

Faiture of boundedness for polarized lc good minimal models

When X is a **good minimal model**, that is, K_X is semiample, there exists a CY fibration $f\colon X \to Z$ to a normal variety Z with $\dim Z = \kappa(X)$ and a canonical bundle formula

$$K_X \sim_{\mathbb{Q}} f^*(K_Z + B_Z + M_Z)$$

such that $K_Z + B_Z + M_Z$ is ample. We can then regard $(Z, B_Z + M_Z)$ as a **generalised log canonical (glc) model**.

Theorem (Hacon–J. 2025)

*There exist projective **lc** minimal **surfaces** X with $\kappa(X) \in \{0, 1, 2\}$ and ample $\mathbb{Q}$-Cartier **Weil** divisors A such that $\mathrm{vol}(K_X + A)$ is fixed, but the Cartier index of A is unbounded. Moreover, X is not bounded.*

Polarized klt good minimal models

Motivation. As a consequence of the **unboundedness** of polarized **lc** good minimal models, constructing **compact moduli spaces** for polarized Calabi–Yau varieties (and more generally good minimal models) becomes significantly more difficult.

Setup. We restrict to **polarized klt good minimal models** (X, A):

- X is **klt**,
- K_X **semiample**, defining the Iitaka fibration $f \colon X \to Z$ with general fiber F,
- A a $\mathbb{Q}$-Cartier **Weil divisor** on X, **ample over** Z.

Key input (Birkar 2021, 2023).

- **Bases**: generalized log canonical models $(Z, B_Z + M_Z)$ are bounded
- **Fibers**: polarized Calabi–Yau varieties $(F, A|_F)$ are bounded

Boundedness of polarized klt good minimal models I

Theorem (J.–Jiao–Zhu 2025)

Let $d \in \mathbb{N}$, and $u, v \in \mathbb{Q}^{>0}$. Consider the set of polarized klt good minimal models $(X, A) \to Z$ satisfying the following:

- *X is **klt** of dimension d,*
- *A is **ample over** Z, with $\mathrm{vol}(A|_F) \le u$,*
- $\mathrm{Ivol}(K_X)(= (K_Z + B_Z + M_Z)^{\dim Z}) = v$.

Then the set of such X is boundedness in codimension one.

A set of varieties $\mathcal{P}$ is said to be ***bounded in codimension one*** if there exist a projective morphism $\pi : \mathcal{X} \to S$ between finite type schemes such that for every $X \in \mathcal{P}$ there exists $s \in S$ with X isomorphic to $\mathcal{X}_s$ in codimension one.

Boundedness of polarized klt good minimal models II

Theorem (J. 2023)

Let $d \in \mathbb{N}$, $u \in \mathbb{Q}^{>0}$, and $\sigma \in \mathbb{Q}[t]$ be a polynomial. Consider the set of polarized klt good minimal models $(X, A) \to Z$ satisfying the following:

- *X is **klt** of dimension d,*
- *A is **ample over** Z, with $\mathrm{vol}(A|_F) \leq u$,*
- *$(K_X + tA)^d = \sigma(t)$.*

*Then the set of such X forms a bounded family. Moreover, there exist positive natural numbers m and τ, depending only on (d, u, σ), such that $m(K_X + \tau A)$ is **very ample**.*

We denote this set by $\mathcal{G}_{klt}(d, u, \sigma)$.

Moduli of polarized klt good minimal models

Theorem (J. 2023)

Fix $d \in \mathbb{N}$, $u \in \mathbb{Q}^{>0}$, and $\sigma \in \mathbb{Q}[t]$. The moduli functor $\mathfrak{G}_{klt}(d, u, \sigma)$ for the set $\mathcal{G}_{klt}(d, u, \sigma)$ is a separated Deligne–Mumford stack of finite type, which admits a coarse moduli space $G_{klt}(d, u, \sigma)$ as a separated algebraic space.

We prove that a certain line bundle is ample on $\widetilde{G}_{klt}(d, u, \sigma)$.

Theorem (J. 2026)

Let $\delta : \widetilde{G}_{klt}(d, u, \sigma) \to G_{klt}(d, u, \sigma)$ be the normalization of $G_{klt}(d, u, \sigma)$, then $\widetilde{G}_{klt}(d, u, \sigma)$ is a quasi-projective scheme. Moreover, if the non-normal locus of $G_{klt}(d, u, \sigma)$ is proper, then $G_{klt}(d, u, \sigma)$ is a quasi-projective scheme.

There exists an example due to Kollár of a line bundle L on a non-normal, non-proper space X such that L is not ample, whereas its pullback to the normalization $\widetilde{X}$ is ample. Therefore, the quasi-projectivity of $\widetilde{G}_{klt}(d, u, \sigma)$ does not imply that of $G_{klt}(d, u, \sigma)$.

Weak positivity

The quasi-projectivity result is based on Viehweg's ampleness criterion in the non-proper setting and the following weak positivity (a non-proper analogue of nefness) result.

Theorem (J. 2026)

Let $f\colon X \to S$ be a locally stable family of klt varieties over a reduced quasi-projective scheme S such that K_X is f-semiample. Let L be an f-semiample line bundle on X. Assume that:

- *there exists $l \in \mathbb{N}$ such that $\mathrm{lct}(X_s, |L_s|) > \frac{1}{l}$ for all $s \in S$, and*
- *the sheaf f_*L is locally free of rank $r > 0$ and compatible with arbitrary base change.*

Then for every natural number $q \geq 2$ such that qK_X is Cartier, the sheaf

$$\mathcal{E}_{q,l} := \Big(\bigotimes^{r} f_*(\mathcal{O}_X(qlK_{X/S}) \otimes L^q)\Big) \otimes \det(f_*L)^{-q}$$

is weakly positive over S.

Thank you!

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Questions are welcome.